

SAMEER RAI

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SUMMARY

Computer Engineering student and LeetCode **Knight** (Top **4.8%**, **1895** rating, **470+** problems solved) with hands-on experience building scalable backend systems and REST APIs in Java, Spring Boot, and Node.js/Express, backed by PostgreSQL and MongoDB. Comfortable across the stack – from concurrent service architecture and API security to React frontends – with a focus on system design, data structures, and production-ready code.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, Punjab
BTech in Computer Engineering
St. Joseph Montessori School, Lucknow, Uttar Pradesh
ISC: **96.00%** | ICSE: **95.60%**

Aug 2023 – June 2027
CGPA: **8.59/10**
Mar 2015 – Feb 2022

EXPERIENCE

Project Intern, RAO Quantum Technologies

Jun 2026 – Jul 2026

Node.js, Express.js, Twilio Programmable Messaging, WhatsApp Cloud API, Webhooks

- Migrated a production WhatsApp ordering bot from Meta Cloud API to Twilio Programmable Messaging in a live Node.js/Express codebase, isolating the change to **3 of 9 files** in the existing module, with zero modification to the core session state machine.
- Reverse-engineered a 3-layer (domain/application/infrastructure) architecture spanning 2 parallel WhatsApp flows before implementation, surfacing a pre-existing architectural gap for the team.
- Designed a template-free interactive-message adapter to replace unsupported WhatsApp list/button UI, and implemented **HMAC-based webhook signature verification** for inbound requests.
- Validated the migration with **75/75 passing automated tests** (24 unit, 51 integration) and shipped a credential-less dry-run mode with a scoped patch package for drop-in integration.

PROJECTS

API Monitoring Dashboard

GitHub | Live Demo
Feb 2026

Java, Spring Boot, WebClient, PostgreSQL, React.js, Docker

- Built a self-hosted, multi-user API uptime/latency monitoring platform with JWT-authenticated, per-user data isolation enforced at the repository layer – a request for another user's service **404s instead of 403s**, never confirming it exists.
- Engineered a non-blocking health-check engine (Spring WebClient, concurrency of 8) with exponential-backoff retry, polling due services in parallel on a 15-second tick – benchmarked at **5-6x faster** than sequential polling across 20 services.
- Hardened submitted URLs against SSRF via real DNS-resolution checks against private/loopback/reserved ranges, and surfaced **30-day windowed** historical analytics (uptime %, latency trends, outage/recovery detection) on a live dashboard.

Disaster Intelligence System

GitHub
Dec 2025

Python, FastAPI, Scikit-learn, Gemini API, REST APIs, React.js

- Designed a multimodal ML pipeline combining BiLSTM (GloVe) for text filtering and ResNet-50 for image classification, achieving **93% image accuracy** and **82% text accuracy**.
- Processed **10K+ tweets** and **13K+ images** with a two-stage conditional pipeline, reducing unnecessary computation and enabling real-time disaster detection and alerting.
- Integrated FastAPI inference, automated report generation, and email-based alerting for end-to-end deployment.

TOPSIS Decision Analysis Package

GitHub | PyPI
Jan 2026

Python, NumPy, Pandas, PyPI, Flask

- Published Topsis-Sameer-102303773 on PyPI, a pip-installable Python package implementing the TOPSIS algorithm for multi-criteria decision making.
- Built CLI utilities and a Flask-based web interface to automate ranking of alternatives from CSV datasets.

ACHIEVEMENTS

- **LeetCode Knight** (Top 4.8%) – Rating **1895**; 470+ problems solved; ranked **1,254/38,791** in Weekly Contest 509.
- **Code Marathon** (Link) – Secured **1st Runner-Up** among 80+ teams in a relay-style coding competition.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, JavaScript, SQL

Frameworks: Spring Boot, React.js, FastAPI, Flask, Node.js, Express.js, Scikit-learn

Databases: MySQL, MongoDB, PostgreSQL

Developer Tools: Git, Docker, Linux (Ubuntu), VS Code, PyCharm, Google Colab

Concepts: Data Structures & Algorithms, System Design, Microservices, REST API Design, OOP

CERTIFICATES AND SHORT COURSES

- **Unit Testing in Java** (Link): JUnit, Service Testing.
- **CI/CD Pipeline with Docker: From Code to Deployment** (Link): Docker, CI/CD Pipelines.
- **Working with BigQuery** (Link): SQL, Big Data, Data Visualization.